

INTERNSHIP REPORT
ON
SMART TEMPERATURE
MONITOR
(TASK-1)
IN THE DOMAIN OF
EMBEDDED SYSTEMS AND IOT
BY
MANOGYNA.R
AT
MAINCRAFTS TECHNOLOGY
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INTRODUCTION SECTION

- **Introduction to Embedded System**

Embedded systems are specialized computing systems designed to perform specific tasks within a larger system. They consist of a combination of hardware and software, where a microcontroller or microprocessor is programmed to carry out dedicated functions.

Unlike general-purpose computers, embedded systems are task-oriented and work with limited resources such as memory, processing power, and energy.

Embedded systems are widely used in various applications such as consumer electronics, automotive systems, industrial machines, medical devices, and home automation systems.

- **Real World Applications**

Embedded systems are widely used in various fields due to their efficiency, reliability, and ability to perform dedicated tasks.

1. Smart Home Systems

Embedded systems are used in smart home devices such as smart lighting, smart thermostats, security systems, and home automation devices. These systems allow users to control appliances remotely and improve energy efficiency.

2. Automotive Systems

Modern vehicles use embedded systems for functions such as engine control, anti-lock braking systems (ABS), airbag control, navigation systems, and parking assistance, making driving safer and more efficient.

3. Healthcare Devices

Embedded systems are used in medical equipment such as patient monitoring systems, digital thermometers, ECG machines, and insulin pumps, helping in accurate diagnosis and treatment.

4. Industrial Automation

In industries, embedded systems are used for controlling machines, monitoring processes, and improving production efficiency through automation.

5. Robotics

Embedded systems play a key role in robotics for controlling sensors, motors, and decision-making processes, enabling robots to perform complex tasks.

Microcontrollers VS Microprocessor

Microcontroller	Microprocessor
Designed for specific control-based applications	Designed for general-purpose computing tasks
Has built-in memory and input/output interfaces	Requires external memory and peripherals
Optimized for real-time embedded applications	Optimized for complex processing tasks
Used in compact electronic products	Used in desktops, laptops, and servers

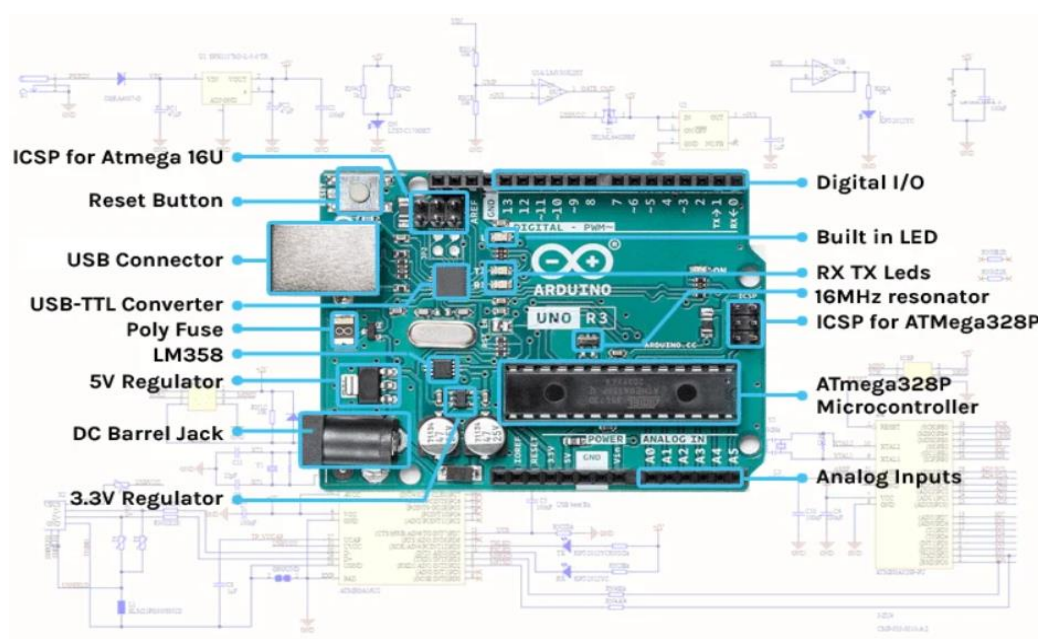
COMPONENT STUDY

Microcontrollers are the core components of embedded systems that help in controlling and processing different operations. They are compact integrated circuits designed to execute specific tasks efficiently. In this different microcontroller platforms such as Arduino, ESP32, and Raspberry Pi Pico are studied to understand their features, working, and applications in Embedded Systems and IoT.

1. Arduino

Arduino is an open-source microcontroller platform widely used for embedded system and IoT projects. It is easy to program and interface with sensors, LEDs, motors, and other electronic components. Arduino boards are mainly used for automation, monitoring, and control applications. It is beginner-friendly and supports programming through the Arduino IDE.

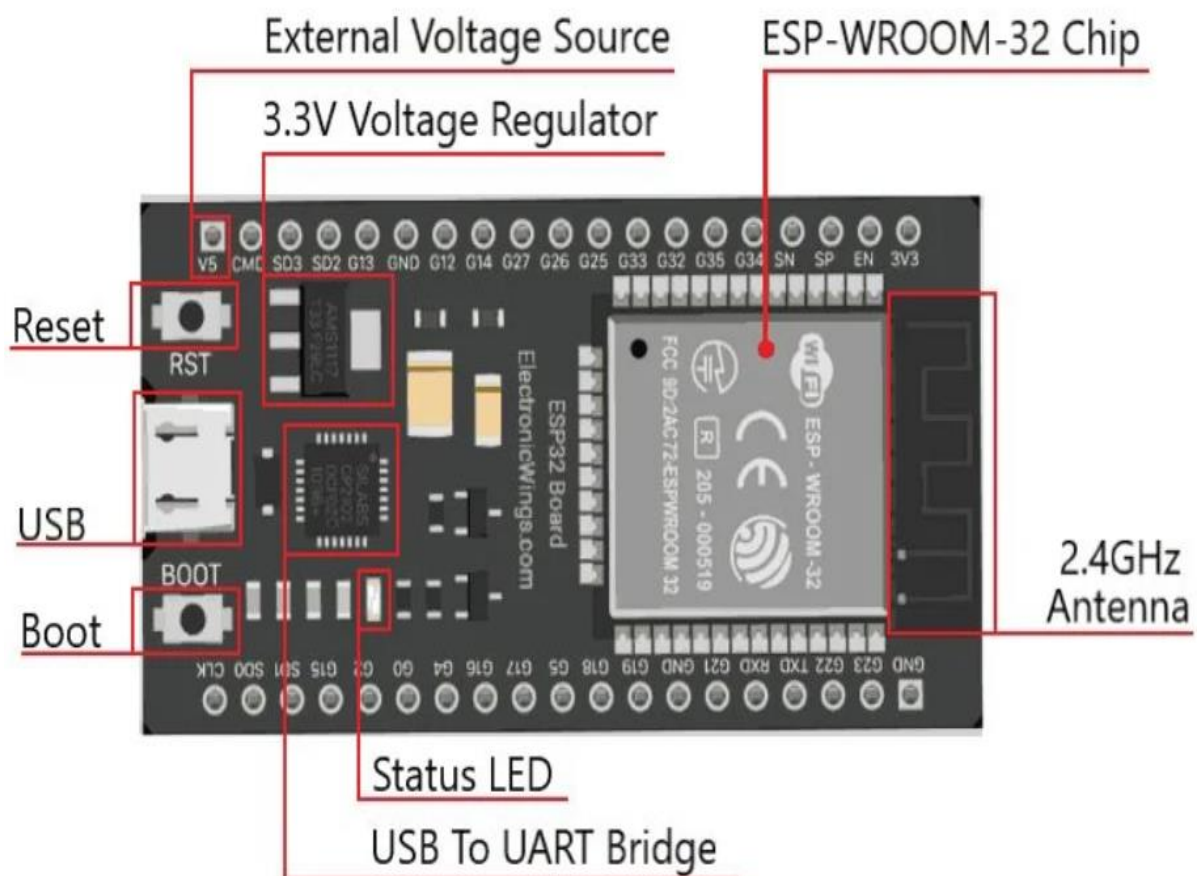
Arduino acts as the control unit that processes input data from sensors and produces the required output based on programmed instructions.



2. ESP32

ESP32 is a powerful microcontroller developed for IoT applications. It has built-in Wi-Fi and Bluetooth connectivity, making it suitable for wireless communication and smart devices. ESP32 provides high processing speed, low power consumption, and support for real-time applications. ESP32 also supports dual-core processing, which allows it to perform multiple tasks efficiently at the same time.

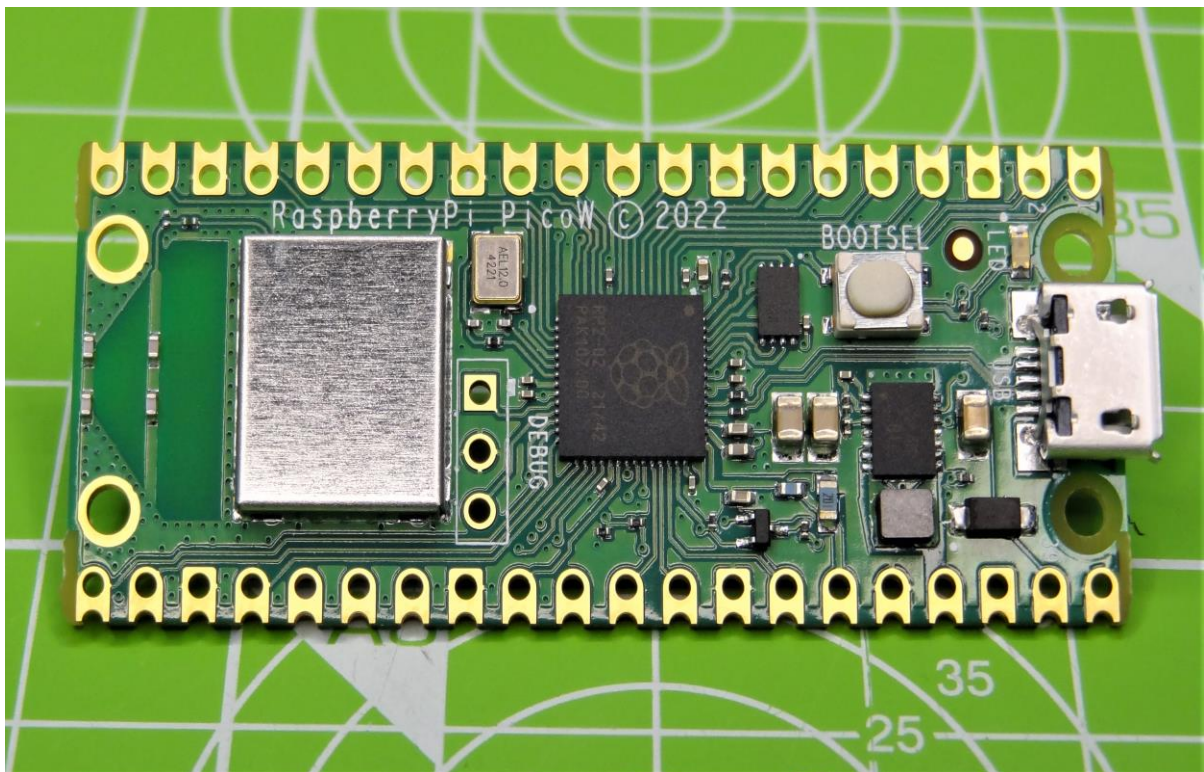
ESP32 is mainly used in IoT-based systems for wireless monitoring, remote control, and cloud connectivity applications.



3. Raspberry Pi Pico

Raspberry Pi Pico is a low-cost microcontroller board developed by Raspberry Pi. It is based on the RP2040 microcontroller chip and is designed for embedded and automation projects. It supports multiple input/output interfaces and can be programmed using Micro Python or C/C++.

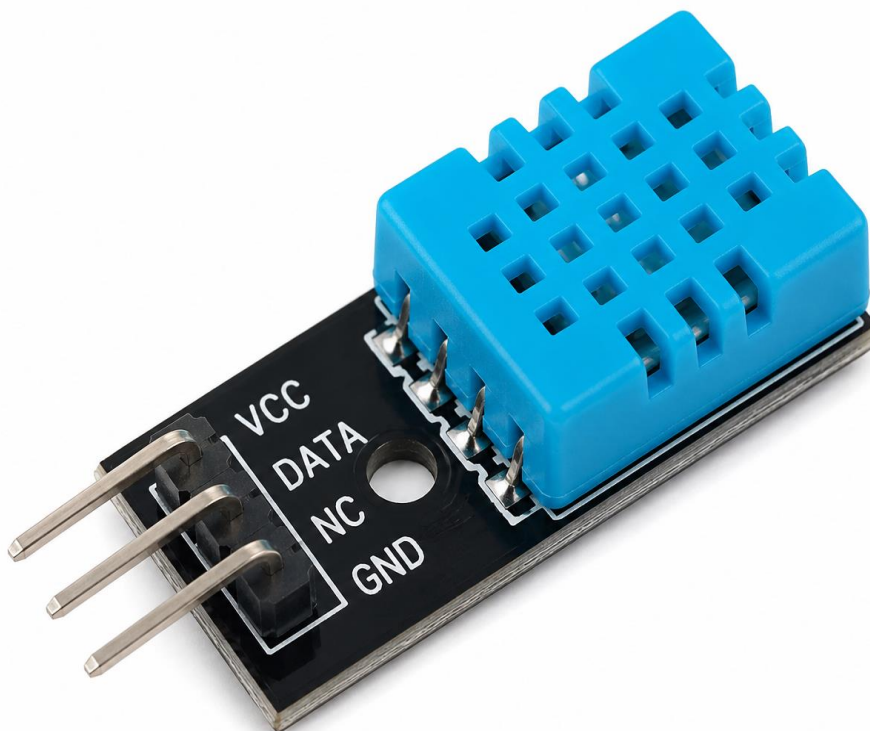
Raspberry Pi Pico is used for controlling electronic devices, sensor interfacing, and real-time embedded applications with efficient performance.



4. SENSORS

Sensors are electronic devices that detect physical changes in the environment and convert them into electrical signals that can be processed by microcontrollers such as ESP32 or Arduino. Sensors play a major role in Embedded Systems and Internet of Things (IoT) applications because they help systems collect real-time information from the surroundings. These devices improve automation, monitoring, accuracy, and efficiency in smart systems.

In modern technology, sensors are widely used in smart homes, industrial automation, healthcare systems, security systems, weather monitoring stations, and IoT-based projects. Sensors can measure different environmental parameters such as temperature, light intensity, pressure, humidity, motion, and gas concentration. The data collected from sensors helps microcontrollers make decisions and perform actions automatically.



Types of Sensors:

(i) Temperature Sensor

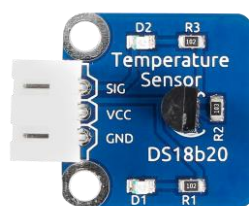
A temperature sensor is an electronic device used to measure temperature and convert it into electrical signals that can be processed by a microcontroller or monitoring system. It helps in detecting changes in environmental temperature and provides accurate readings for various applications. Temperature sensors are widely used in embedded systems, industrial automation, healthcare devices, weather monitoring systems, home automation, and IoT applications.

Working Principle

The working principle of a temperature sensor is based on detecting temperature changes and converting them into electrical signals. When the surrounding temperature changes, the properties of the sensing element inside the sensor also change. The processed temperature value can then be displayed on screens or used to activate alarms, cooling systems, or other control devices. Some sensors provide analog output while others provide digital output depending on their design.

Applications

- Industrial temperature monitoring systems.
- Medical and healthcare devices.
- Smart home automation systems.



(ii) Motion Sensor

A motion sensor is an electronic device used to detect movement or motion within a specific area. It is commonly used in security systems, automation systems, robotics, and smart devices. Motion sensors help systems respond automatically when movement is detected, improving safety, energy efficiency, and automation. These sensors are widely used in homes, industries, offices, and public places for automatic control applications.

Working Principle

The motion sensor works by detecting movement in its surrounding area and converting it into electrical signals. When a person or object moves within the sensor range, the sensor identifies the change and sends a signal to the control system. The microcontroller then processes the signal and performs specific actions such as switching ON lights, activating alarms, or controlling devices automatically. Different types of motion sensors use technologies such as infrared radiation, ultrasonic waves, or microwave signals for motion detection.

Applications

- Security and surveillance systems.
- Automatic doors and gates.
- Smart lighting systems.



(iii) Light Sensor

A light sensor is an electronic device used to detect the intensity of light in the surrounding environment. It converts light energy into electrical signals that can be processed by electronic systems. Light sensors are widely used in automatic lighting systems, consumer electronics, industrial automation, and smart devices. These sensors help in improving energy efficiency and enable automatic control based on environmental lighting conditions.

Working Principle

The working principle of a light sensor is based on detecting the amount of light falling on its surface. When the light intensity changes, the electrical properties of the sensor also change. This change is converted into electrical signals and sent to the microcontroller for processing. Based on the received light data, the system can automatically adjust brightness levels, switch lights ON or OFF, or perform other automated operations.

Applications

- Automatic street lighting systems.
- Mobile phone brightness control.
- Solar tracking systems.



PROJECT: SMART TEMPERATURE MONITOR

Objective:

The objective of the Smart Temperature Monitor System is to measure and monitor temperature in real time using a temperature sensor and microcontroller. The system also provides alerts when the temperature exceeds a predefined limit.

Working Principle:

The Smart Temperature Monitor System works by continuously sensing and monitoring the surrounding temperature using a temperature sensor and microcontroller. The system processes the temperature data in real time and provides alerts whenever the temperature exceeds the predefined limit. The complete working process of the system is explained in the following stages.

1. Temperature Detection

The temperature sensor detects the surrounding temperature from the environment.

2. Signal Conversion

The detected temperature is converted into electrical signals by the sensor.

3. Data Transmission

The sensor sends the generated signals to the microcontroller.

4. Data Processing

The microcontroller reads and processes the temperature data.

5. Temperature Display

The processed temperature value is displayed on the output device such as LCD or Serial Monitor.

6. Threshold Comparison

The microcontroller compares the measured temperature with the predefined threshold value.

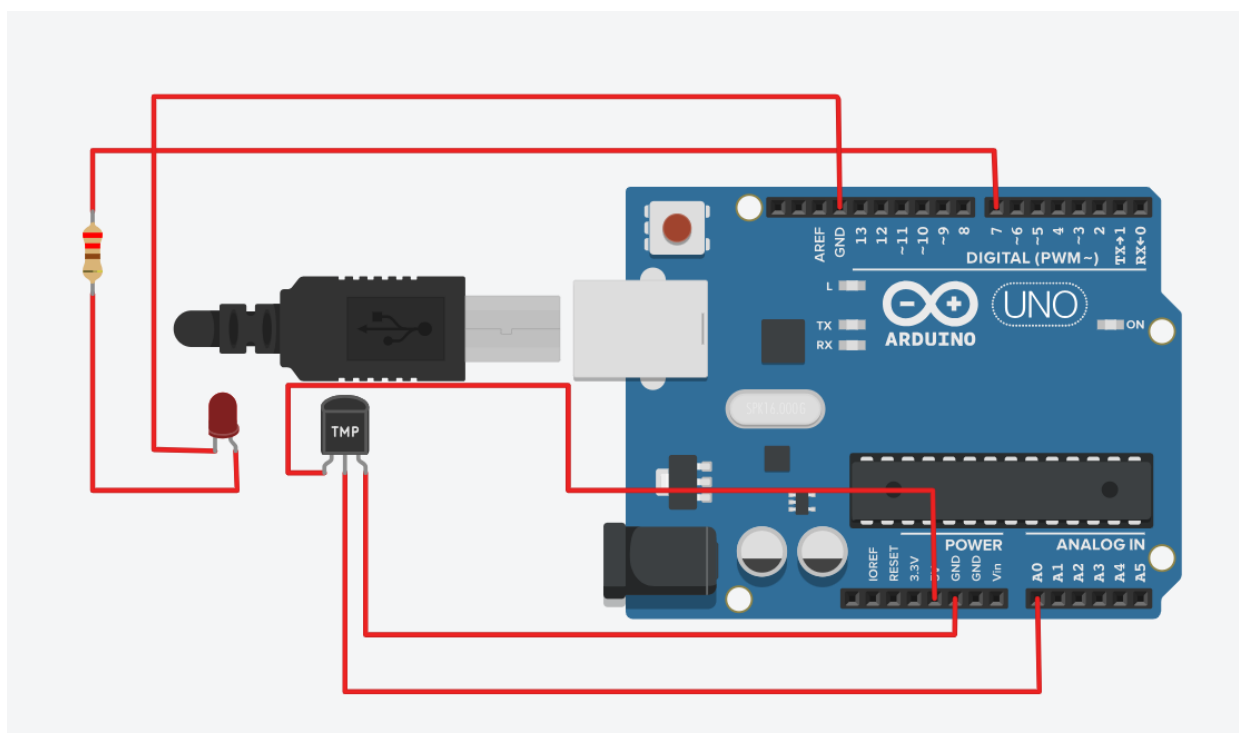
7. Alert Generation

If the temperature exceeds the set limit, the system activates an LED or buzzer alert.

8. Continuous Monitoring

The system continuously repeats the process for real-time temperature monitoring.

CIRCUIT DIAGRAM



CODE:

```
int ledPin = 7;
int tempPin = A0;
void setup()
{
    pinMode(ledPin,OUTPUT);
    Serial.begin(9600); //Setting the serial communication between the Arduino
and Serial monitor at 9600 bits per second
}
void loop ()
{
    int result = analogRead(tempPin); //Reads the temperature value from the
Arduino analog input pin A0
    Serial.println(result);
    if(result>300)
    {
        digitalWrite(ledPin,HIGH);
    }
    else
    {
        digitalWrite(ledPin,LOW);
    }
    delay (100);
}
```

CODE EXPLANATION:

1. **int ledPin = 7;**

This line declares an integer variable named ledPin and assigns pin number 7 for connecting the LED.

2. **int tempPin = A0;**

This line declares an integer variable named tempPin and assigns the analog pin A0 for connecting the temperature sensor.

3. **void setup ();**

The setup() function is used to initialize the system settings. It runs only once when the Arduino board is powered ON or reset.

4. **pinMode(ledPin, OUTPUT);**

This line sets the LED pin as an output pin so that the Arduino can control the LED.

5. **Serial.begin(9600);**

This line starts serial communication between the Arduino and the Serial Monitor at a baud rate of 9600 bits per second.

6. **void loop ()**

The loop () function contains the main program instructions and runs continuously.

7. **int result = analogRead(tempPin);**

This line reads the analog value from the temperature sensor connected to pin A0 and stores it in the variable result.

8. Serial.println(result);

This line displays the temperature sensor value on the Serial Monitor.

9. if (result>300)

This condition checks whether the temperature sensor value is greater than 300.

10. digitalWrite(ledPin, HIGH);

If the temperature value is greater than 300, this line turns ON the LED.

11. Else

The else statement executes when the condition result > 300 is false.

12. digitalWrite(ledPin,LOW);

This line turns OFF the LED when the temperature value is less than or equal to 300.

13. delay (100);

This line pauses the program execution for 100 milliseconds before repeating the loop.

REAL – WORLD APPLICATIONS

1. Industrial Monitoring:

The system is used in industries to continuously monitor machine and equipment temperature to prevent overheating and damage.

2. Healthcare Applications

Smart temperature monitoring systems are used in hospitals and laboratories to monitor body temperature, medical equipment, and storage conditions for medicines.

3. Home Automation

The system can be used in smart homes to control fans, air conditioners, and other appliances based on temperature conditions.

4. Agricultural Monitoring

It is used in agriculture and greenhouses to maintain suitable temperature conditions for plant growth.

5. Server and Computer Rooms

The system helps monitor temperature in server rooms and data centres to prevent system failures caused by overheating.

6. Weather Monitoring Systems

Temperature monitoring systems are used in weather stations to measure and record environmental temperature changes.

Project Video link:

"C:\Users\rmano\Videos\Captures\Circuit design Dazzling Gogo-
Kup - Tinkercad - Google Chrome 2026-05-19 16-23-35.mp4"